

## **SET ANSWER 2**

### **Scenario based answer**

#### **Scenario 1: Forecast for Next Week**

**Logic:**

1. Arrange website traffic data in chronological order.
2. Split the data into training and testing sets.
3. Check whether there is a trend or seasonality.
4. Choose a suitable time-series model.
5. Train the model using historical data.
6. Forecast the next 7 days.
7. Compare the forecast with actual values when they become available.

#### **Scenario 2: Choosing a Time Series Model**

**Logic:**

- A simple model such as Moving Average or Simple Exponential Smoothing can be used.
- If the series is stationary, ARMA/ARIMA with suitable parameters can also be considered.

#### **Scenario 3: Monthly Sales Showing Steady Growth**

**Logic:**

- Since sales are steadily increasing, choose a model that can capture trend.
- Holt's Linear Trend method (Double Exponential Smoothing) is a suitable choice.
- Train the model on historical sales and forecast future months.

## **Scenario 4: Periodic Increase Every December**

### **Logic:**

- December repeatedly has higher sales.
- The pattern occurs every 12 months.
- The forecasting model should include a seasonal component.
- For example, use Holt-Winters / Exponential Smoothing with yearly seasonality.

## **Scenario 5: Handling Missing Values**

### **Logic:**

1. Find where the missing values occur.
2. Check why they are missing.
3. If the missing values are isolated, use an appropriate method such as:
  - Forward fill
  - Backward fill
  - Interpolation
4. For longer missing periods, investigate carefully before filling.
5. Make sure the time series remains in correct chronological order.

## **Scenario 6: Evaluating Forecast Accuracy**

### **Logic:**

1. Split the historical data into training and testing data.
2. Train the model on the training data.
3. Forecast the test period.
4. Compare predicted values with actual values.
5. Calculate metrics such as:
  - MAE – Mean Absolute Error
  - RMSE – Root Mean Squared Error
  - MAPE – Mean Absolute Percentage Error
6. Lower error generally means better forecasting performance.